Musical examples

The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11



🔊 Click to Play

We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
  [{a ⟦I None⟧}, {a ⟦IV None⟧}]
```

```
Output analysis:
  size=11
  Concatenate
    {"a" ⟦I None⟧}
    {"a" ⟦IV None⟧}
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12



🔊 **Click to Play**

This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
Input surface:
  [{a ⟦I-V None⟧}, {b ⟦I-III None⟧}, {a ⟦VI-II None⟧}, {b ⟦VI-III None⟧}]



Output analysis:
  size=27
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-III None⟧}
      Concatenate
        {"a" ⟦VI-II None⟧}
        {"b" ⟦VI-III None⟧}



  size=27
  Concatenate
    Concatenate
      {"a" ⟦I-V None⟧}
      {"b" ⟦I-III None⟧}
    Concatenate
      {"a" ⟦VI-II None⟧}
      {"b" ⟦VI-III None⟧}
```

Identifying compound themes

example 6-1



🔊 Click to Play

```
Input surface:
  [ {a ⟦I-V None⟧}
  , {b ⟦I-V-I-V Weak⟧}
  , {c ⟦IV-V-I None⟧}
  , {d ⟦VI-VII-I-V Weak⟧}
  , {a ⟦I-V None⟧}
  , {b ⟦I-V-I-V Weak⟧}
  , {c ⟦V None⟧}
  , {e ⟦I Strong⟧} ]




Output analysis:
  size=44
  Concatenate
    Ternary
      Concatenate
        {"a" ⟦I-V None⟧}
        {"b" ⟦I-V-I-V Weak⟧}
      Concatenate
        {"c" ⟦IV-V-I None⟧}
        {"d" ⟦VI-VII-I-V Weak⟧}
    Concatenate
      {"c" ⟦V None⟧}
      {"e" ⟦I Strong⟧}



  size=44
  HybridAntCont
    Ternary
      Concatenate
        {"a" ⟦I-V None⟧}
        {"b" ⟦I-V-I-V Weak⟧}
      Concatenate
        {"c" ⟦IV-V-I None⟧}
        {"d" ⟦VI-VII-I-V Weak⟧}
    Concatenate
      {"c" ⟦V None⟧}
      {"e" ⟦I Strong⟧}
```

example 6-2



🔊 Click to Play

```
Input surface:
  [ {a ⟦I None⟧}
  , {b ⟦I None⟧}
  , {c ⟦I None⟧}
  , {a ⟦V None⟧}
  , {b ⟦V None⟧}
  , {c ⟦V None⟧}
  , {b ⟦I-VI None⟧}
  , {b ⟦IV-II None⟧}
  , {b ⟦V None⟧}
  , {d ⟦I-II-V-I Strong⟧} ]




Output analysis:
  size=62
  Sentence
    basicIdea= "a"+"b"+"c"
    presentation= Concatenate
                    UseIdeaWith   $0 ⟦I None⟧+⟦I None⟧+⟦I None⟧
                    UseIdeaWith   $0 ⟦V None⟧+⟦V None⟧+⟦V None⟧
    continuation= Sentence
                    basicIdea= "b"
                    presentation= Concatenate
                                    UseIdeaWith   $0 ⟦I-VI None⟧
                                    UseIdeaWith   $0 ⟦IV-II None⟧
                    continuation= Concatenate
                                    {$0 ⟦V None⟧}
                                    {"d" ⟦I-II-V-I Strong⟧}



  size=65
  Concatenate
    {"a" ⟦I None⟧}
    Concatenate
      {"b" ⟦I None⟧}
      Concatenate
        {"c" ⟦I None⟧}
        Concatenate
          {"a" ⟦V None⟧}
          Concatenate
            {"b" ⟦V None⟧}
```

```
Concatenate
  {"c" ⟦V None⟧}
  Sentence
    basicIdea= "b"
    presentation= Concatenate
                    UseIdeaWith   $0 ⟦I-VI None⟧
                    UseIdeaWith   $0 ⟦IV-II None⟧
    continuation= Concatenate
                    {$0 ⟦V None⟧}
                    {"d" ⟦I-II-V-I Strong⟧}
```

example 6-5



🔊 Click to Play

```
Input surface:
  [ {a ⟦I-V None⟧}
  , {a ⟦V-I None⟧}
  , {b ⟦IV-I None⟧}
  , {b ⟦IV-I None⟧}
  , {c ⟦II None⟧}
  , {c ⟦I None⟧}
  , {d ⟦V Weak⟧} ]




Output analysis:
  size=40
  Sentence
    basicIdea= "a"
    presentation= Concatenate
                    UseIdeaWith   $0 ⟦I-V None⟧
                    UseIdeaWith   $0 ⟦V-I None⟧
    continuation= Concatenate
                    TwiceFormalUnit
                      {"b" ⟦IV-I None⟧}
                    Concatenate
                      {"c" ⟦II None⟧}
                      Concatenate
                        {"c" ⟦I None⟧}
                        {"d" ⟦V Weak⟧}



  size=39
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"a" ⟦V-I None⟧}
      Concatenate
        TwiceFormalUnit
          {"b" ⟦IV-I None⟧}
        Concatenate
          {"c" ⟦II None⟧}
          Concatenate
```

{"c" ⟦I None⟧}
{"d" ⟦V Weak⟧}

Others

supp 6-1



🔊 Click to Play

example 5-11



🔊 Click to Play